

What is a Constructor?

A **Constructor** is a special method used to initialize objects. It is called automatically at the moment an object is created using the new keyword.

```
public class Main {  
    int x;  
    // Constructor  
    public Main() {  
        x = 5;  
    }  
}
```

Critical Rules

- The name **must match** the Class name exactly.
- It **cannot** have a return type (not even void).
- It is invoked only once per object lifetime.

Parametrized Constructors

You can pass data into constructors to set custom initial values for different object instances.

```
public Main(int year, String name) {  
    modelYear = year;  
    modelName = name;  
}  
  
// Creation:  
Main myCar = new Main(1969, "Mustang");
```

Why Use Them?

- Enforces **Object Integrity**: Objects start with a valid state.
- Reduces boilerplate code after object creation.
- Enables **Constructor Overloading** (multiple ways to create an object).